



Introduction

- Vowels are **shorter and more likely to devoice** in these contexts:
 - when they are **[+high]** (Lehiste 1970, Solé and Ohala 2010)
 - when they are before **C₂[-voice]** (Chen 1970)

Is the short duration of devoiced vowels the cause (H1) or the result (H2) of devoicing?

H1: [+high], C₂[-voice] → Shortening → Devoicing

- Prediction that follows from the gestural overlap account (Jun et al. 1997)

H2: [+high], C₂[-voice] → Devoicing → Shortening

- In Tokyo Japanese, coarticulatory shortening due to C₂[-voi] is categorically greater for [+high] vowels (Tanner et al. 2019) because they devoice in that context

RQ1. Is there a significant interaction of V_[+hi] and C_[-voi]2 on the duration of C_[-voi]1 V? ⇒ If yes, then H2 is supported.

RQ2. Is the likelihood of VD correlated with CV duration? ⇒ If yes, then H1 is supported.

RQ3. Does devoicing mediate context effects on CV duration? ⇒ If yes, then H2 is supported

Methods

Tokens of interest: C₁[-voice] V_[±high] C₂[±voice]

Tokyo Japanese (TJ) (n= 116,876)

- Corpus of Spontaneous Japanese** (Maekawa et al. 2000)
- V devoicing annotated in the corpus
- Long V and geminate C are excluded

Seoul Korean (SK) (n = 50,349)

- Seoul Corpus** (Yun et al. 2015)
- C₂[+voice] = {voiced lenis obstruents, sonorants}
- C₂[-voice] = {voiceless fortis/aspirated obstruents}
- V devoicing, lenis voicing determined by Voice Report in Praat

Baysian mixed-effects regression models

- DV: Vowel devoicing (Bernoulli with logit link), CV duration (Gaussian)
- IV: V height (binary), C2 voicing (binary)
- Covariates: Preceding C manner, Following C manner, speech rate
- by-speaker and by-word random intercepts and random slopes

Mediation analysis:

Compares regression models fit with and without a mediator variable (Cohen Priva and Gleason 2020, Katz and Pitzanti 2019, Lee 2025)

If context *causes* duration shortening mostly via vowel devoicing (H2), then the **indirect effect** on CV duration via vowel devoicing should be **similar to the total effect** of contexts on CV duration

Discussion

TJ: Strong support for H2 (Context → Devoicing → Shortening)

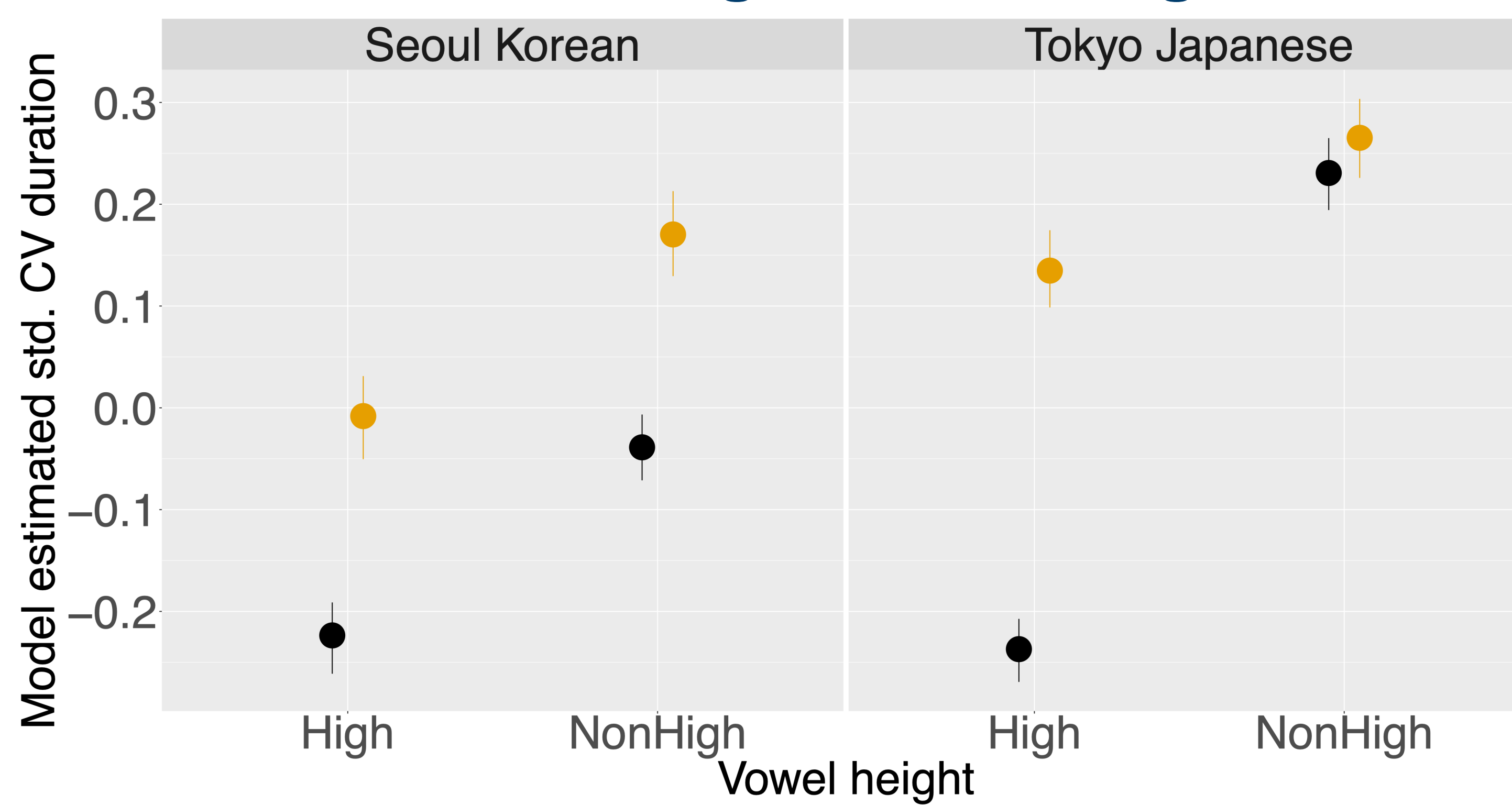
SK: Weak support for H1 (Context → Shortening → Devoicing)

- Potential reason for the difference between SK and TJ: **Phonologization**
- Coarticulation as a precursor to a phonological process (Beddor 2009)
- At the **phonetic** level, VD occurs as a result of coarticulatory shortening (and other factors), as predicted by gestural overlap (H1)
- But once VD is **phonologized**, short duration of CV cannot be attributed to coarticulatory shortening (H2)
- Phonologization may change the causal structure of a process and its associated phonetic variables (H1 → H2)**

Conclusion

- Variation in CV duration** can be explained by general coarticulatory shortening in SK, but **not in TJ**.
- The likelihood of VD** can be explained by coarticulatory shortening in SK to some extent, but **not in TJ**.
- The difference could be due to **phonologization**: VD is phonologized in TJ while VD is at the phonetic level in SK.

RQ1. Height:C2 voicing

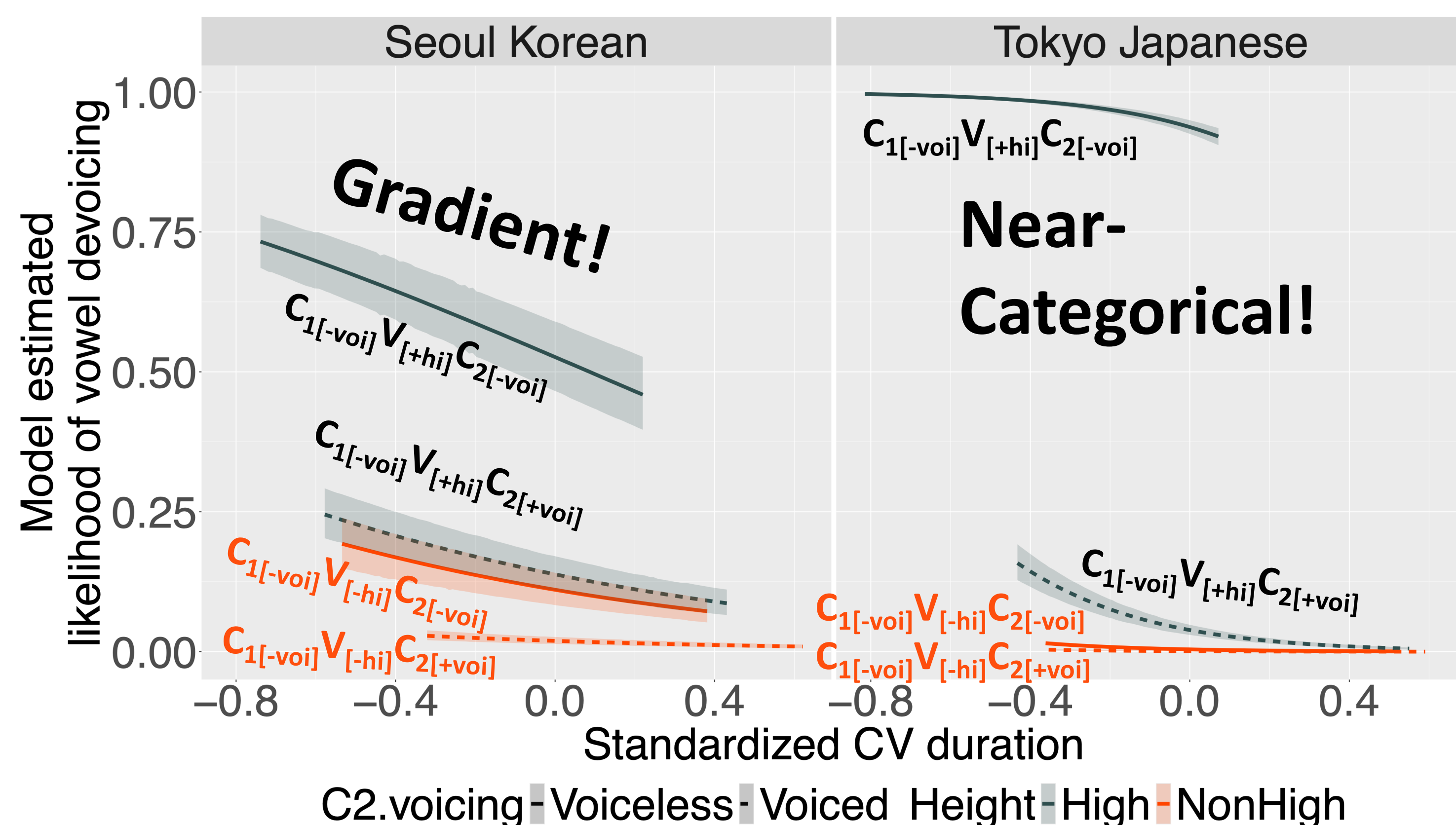


SK: Effect of shortening before C₂[-voi] is **similar** across V heights

TJ: C_[-voi] V_[+hi] becomes much shorter before C₂[-voi] (most of which undergo VD) compared to C_[-voi] V_[-hi] before C₂[-voi]

↳ **Supports H2 for TJ**

RQ2. CV duration x Devoicing



SK: C₁[-voi] V_[+hi] C₂[+voi] ≈ C₁[-voi] V_[-hi] C₂[-voi]

↳ **Supports H1 for SK**

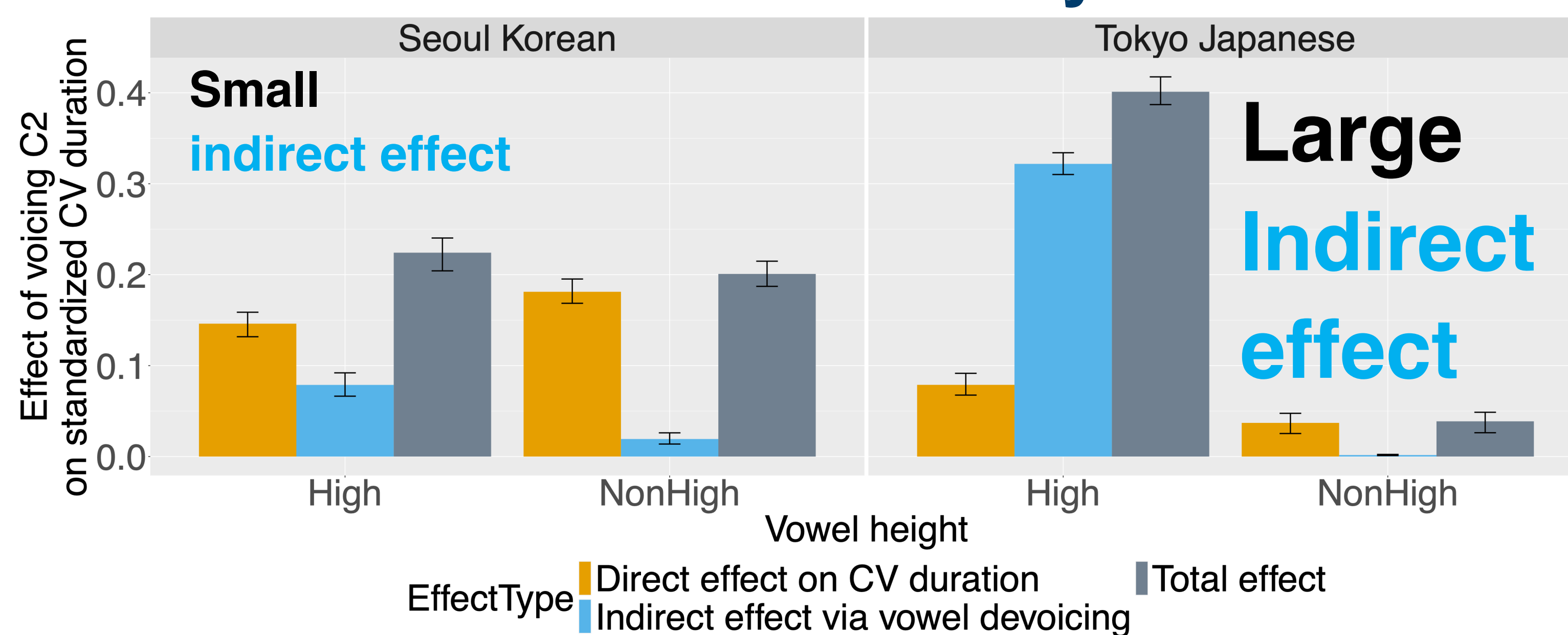
C₁[-voi] V_[+hi] C₂[-voi] > C₁[-voi] V_[-hi] C₂[-voi]

↳ **Challenges H1 for SK**

TJ: Likelihood of VD does not change much by CV duration

↳ **Challenges H1 for TJ**

RQ3. Mediation analysis



SK: The indirect effect via vowel devoicing is **smaller** than the direct effect of C2 voicing on CV duration

↳ **Challenges H2 for SK**

TJ: The indirect effect of C2 voicing on CV duration via vowel devoicing is **similar** to the total effect of C2 voicing on CV duration

↳ Devoicing **mediates** context effects on CV duration

↳ **Supports H2 for TJ**